

Probabilistic Risk Assessment for Power Grids

ECMI 2026

Who are we



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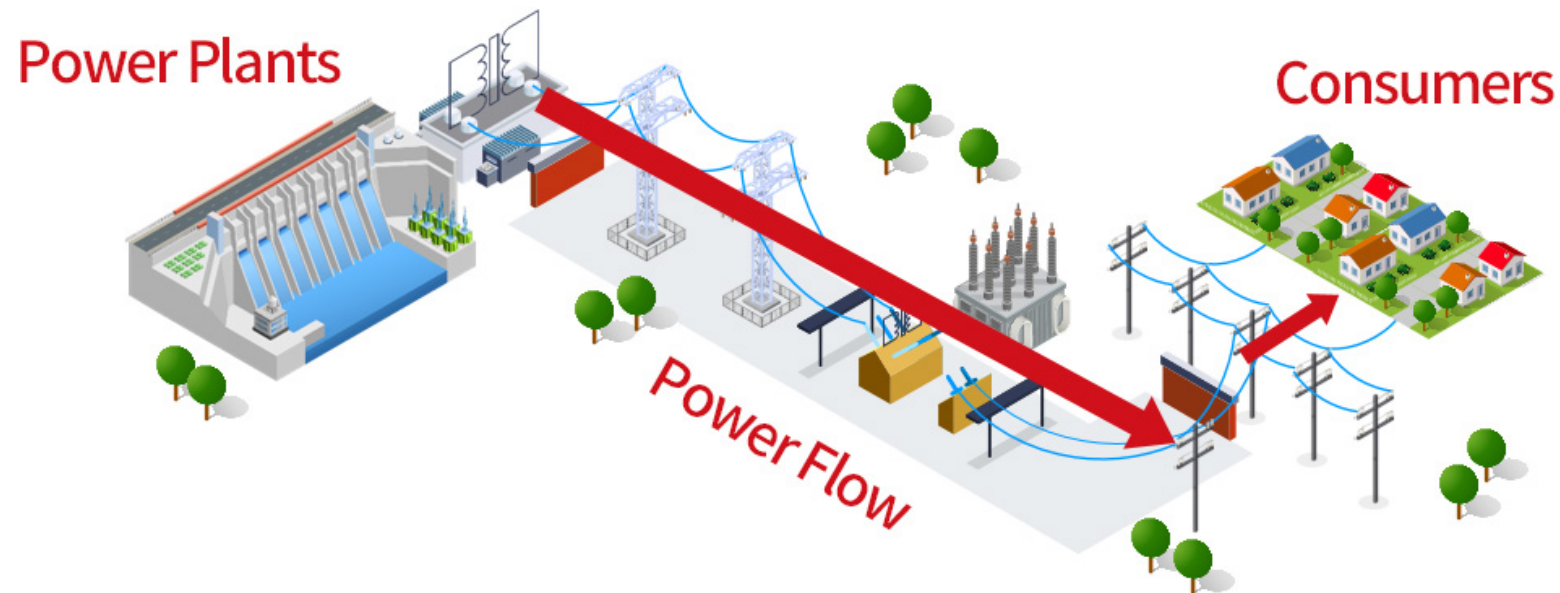


ECMI 2022, Verona

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Power grids



- Power is transported over power lines, between different buses
- Some power is lost (dissipated) due to heating up of the lines
- Delivering power is not always possible: if too much power is lost in transmission, there is not enough for consumers (black-outs)
- Power flow needs to be solvable, based on the supply of power plants and the demand of consumers
- Should also be true when a power line fails => N-1 contingency analysis

ENTSO-E

European Network of Transmission System Operators for Electricity



N-1
contingency analysis

+

PRA
Probabilistic Risk Assessment

Outline and material

Material:

- EU documentation + intro document to PRA
- Matpower: Matlab package for power grids
- Grid data: grid with 200 buses, 8670 data points

Suggested agenda:

- Day 1: N-1 contingency analysis
- Day 2: PRA for expected line failure and blackout, and/or expected load shedding cost using MLS
- Day 3-5: Get creative!

Enjoy!

